CS 301: Distributed Systems & Cloud Infrastructure

Course Syllabus - Fall 2026 | Instructor: Dr. Marcus Vance

1. Course Description

This course provides an in-depth study of distributed systems, fault tolerance, consensus, scalability, and cloud infrastructure architectures. Topics include RPCs, logical clocks, state machine replication, Paxos and Raft algorithms, and distributed storage engines.

2. Course Logistics & Communication

Lectures: Tuesdays & Thursdays 10:00 AM - 11:30 AM (Room EE-104)
Office Hours: Wednesdays 2:00 PM - 4:00 PM (Zoom & Gates Hall 412)
Primary Canvas LMS page contains announcements, grades, and discussion forums.

3. Grading Breakdown

Programming Projects (Labs 1-4): 45%
Midterm Examination: 20%
Final Project (Distributed Key-Value Engine): 25%
Class Participation & Reading Discussions: 10%

4. Late Policy & Academic Honesty

Each student has 3 late days total across all labs without penalty.
All submitted code must be original individual work adhering to university ethics.